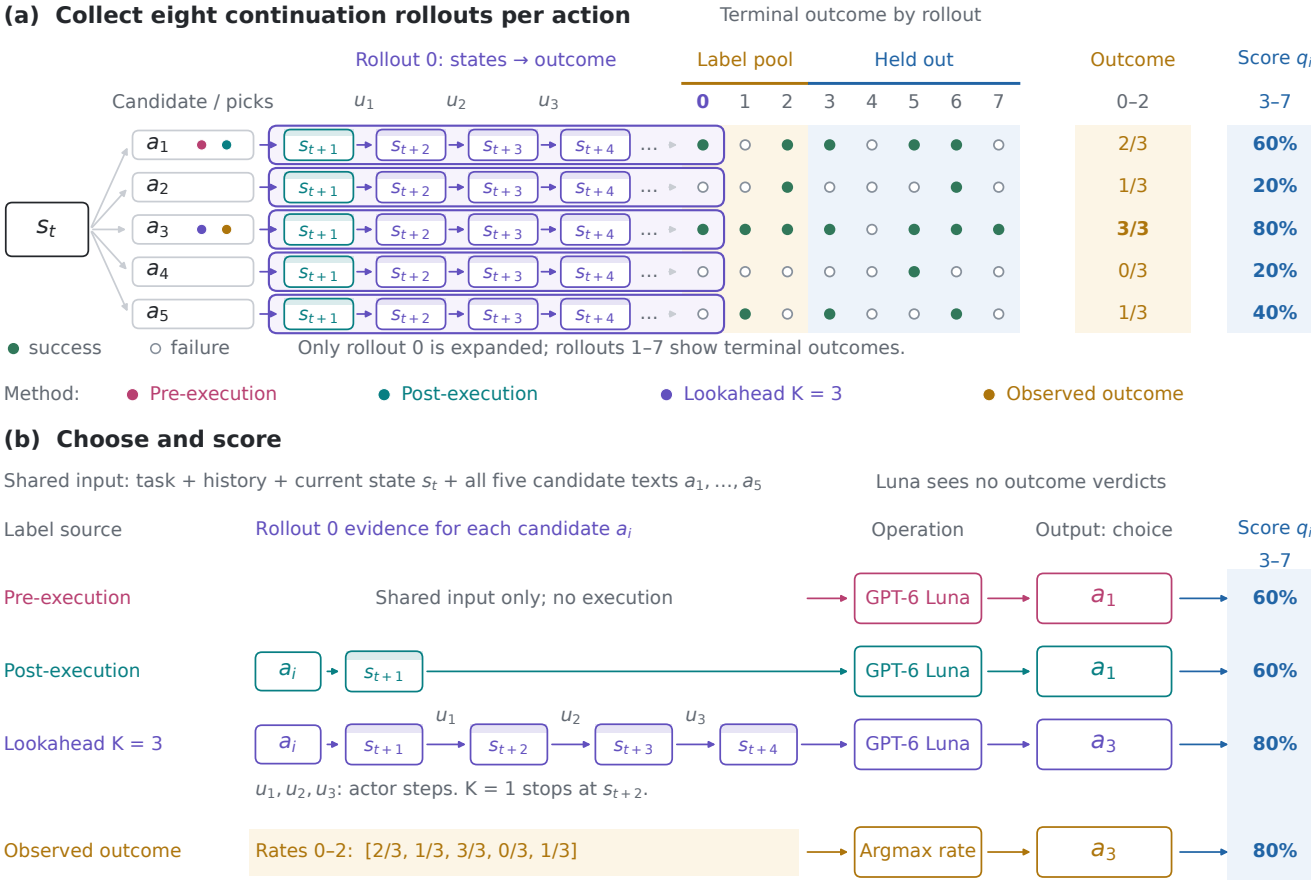
(a) Collect eight continuation rollouts per action
Terminal outcome by rollout
Rollout 0: states → outcome
Label pool
Held out
Outcome
Score q_i
Candidate / picks
u_1 u_2 u_3
0 1 2 3 4 5 6 7
0–2
3–7
s_t
a_1 s_{t+1} s_{t+2} s_{t+3} s_{t+4} … ● ○ ● ○ ● ● ○ 2/3 60%
a_2 s_{t+1} s_{t+2} s_{t+3} s_{t+4} … ○ ○ ● ○ ○ ○ ● ○ 1/3 20%
a_3 s_{t+1} s_{t+2} s_{t+3} s_{t+4} … ● ● ● ● ○ ● ● ● 3/3 80%
a_4 s_{t+1} s_{t+2} s_{t+3} s_{t+4} … ○ ○ ○ ○ ● ○ ○ 0/3 20%
a_5 s_{t+1} s_{t+2} s_{t+3} s_{t+4} … ● ○ ● ○ ○ ● ○ 1/3 40%
● success ○ failure
Only rollout 0 is expanded; rollouts 1–7 show terminal outcomes.
Method: Pre-execution Post-execution Lookahead K = 3 Observed outcome
(b) Choose and score
Shared input: task + history + current state s_t + all five candidate texts $a_1, \ldots, a_5$
Luna sees no outcome verdicts
Label source
Rollout 0 evidence for each candidate a_i
Operation
Output: choice
Score q_i
3–7
Pre-execution
Shared input only; no execution
GPT-6 Luna
a_1
60%
Post-execution
a_i s_{t+1}
GPT-6 Luna
a_1
60%
Lookahead K = 3
a_i s_{t+1} u_1 s_{t+2} u_2 s_{t+3} u_3 s_{t+4}
GPT-6 Luna
a_3
80%
u_1, u_2, u_3: actor steps. K = 1 stops at s_{t+2}.
Observed outcome
Rates 0–2: [2/3, 1/3, 3/3, 0/3, 1/3]
Argmax rate
a_3
80%
Illustrative values · valid rollouts only. Teacher inputs omit outcome verdicts; rollouts 3–7 are reserved for scoring.